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A Leap Towards World Class Education

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Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
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Q1) Define an algorithm. List any four Characteristics of an algorithm.

→ An algorithm is a finite Sequence of well defined, step-by-step instruction used to Solve a specific problem or Perform a Computation.

Characteristics:-

1. Input :- An algorithm takes zero or more inputs.
2. Output :- It Produces at least one Output
3. Definiteness :- Each step is clear, Precise unambiguous.
4. Finiteness :- The algorithm terminates after a finite number of steps

Q2> Define data structure and Abstract data Type (ADT). Give one example of each.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

→ A data structure is a way of organizing, storing, and managing data in memory so that it can be accessed and modified efficiently.

An Abstract Data Type (ADT) is a logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples:
- Data structure: Array
- ADT: Stack (operations: push, pop, ~~peek~~ peek).

Q3) Explain the process of converting a problem into a program. Illustrate the role of algorithms in this process.

→ 1. Problem Definition: Understand requirements and constraints.
2. Analysis: Identify inputs, outputs, and processing.
3. Algorithm Design: Create step-by-step logic to solve the problem.
4. Flowchart/Pseudocode: Represent the algorithm visually or textually.
5. Coding: Convert algorithm into a program using a programming language.
6. Testing & Debugging: Check correctness

and remove errors.

7. Documentation & maintenance: Improve and ~~maint~~ maintain the program.

Role of Algorithm:
The algorithm acts as a blueprint of the solution. It ensures correctness, efficiency, and makes coding easier and error-free.

Q4) Explain linear and non-linear data structures with suitable examples.

→ Linear Data Structure:
Data elements are arranged sequentially.
- Array
- Stack
- Queue
- Linked List

Non-Linear Data Structure:
Data elements are arranged in a hierarchical or interconnected manner

Example :- • Tree
• Graph

Q5) Differentiate b/w algorithm and program. Analy their roles in SDLC.

→

Algorithm	Program
Step-by-step logic	Written code
Language independent	Language dependent
Design level	Implementation level
No syntax rules	follows syntax rule

Role in SDLC:
- Algorithm: Used in design phase to plan the solution
- Program: Used in implementation phase to execute the solution on a computer.

Q6) Examine the relationship ~~at~~ among data, data structure, and algorithm. justify how the choice of data structure affects algorithm efficiency.

→ Relationship:
- Data is raw information.
- Data structure organizes the data.
- Algorithm processes the data using data structures.

Justification:
The efficiency of an algorithm depends on the data structure used.

- Searching is an array → slower (O(n))
- Searching in a binary search tree → faster (O(log n)).

Conclusion:
Choosing the right data structure improves time and space efficiency of Algorithms